

37. DEVELOPMENT OF THE LIVER & GALLBLADDER

DURATION : 08:37

STUDY SHEET

COURSE SUMMARY

In-depth analysis of liver development in its endodermal context, demonstrating the influence of the notochord and the brain. Highlighting inter-intestinal relationships, primitive vascular congestion, and the structuring of the bile ducts. The liver is presented as a central fulcrum in embryological organisation, essential for detoxification. The study of venous and arterial blood flow provides a fundamental perspective on hepatic functions and their spatial regeneration.

LIVER AND GALLBLADDER DEVELOPMENT

The development of the **liver** occurs within an **endodermal** context. The initial impulse at the hepatic level originates from the **notochord**, which influences the formation of the **neural tube**. **Mesenchymal** cells contribute to the formation of the pericardial and diaphragmatic sacs.

The intestine divides into three primitive parts: the **foregut**, **midgut**, and **hindgut**. The junction between the foregut and the beginning of the midgut is crucial for the development of the liver region. These endodermal cells play a key role in this process.

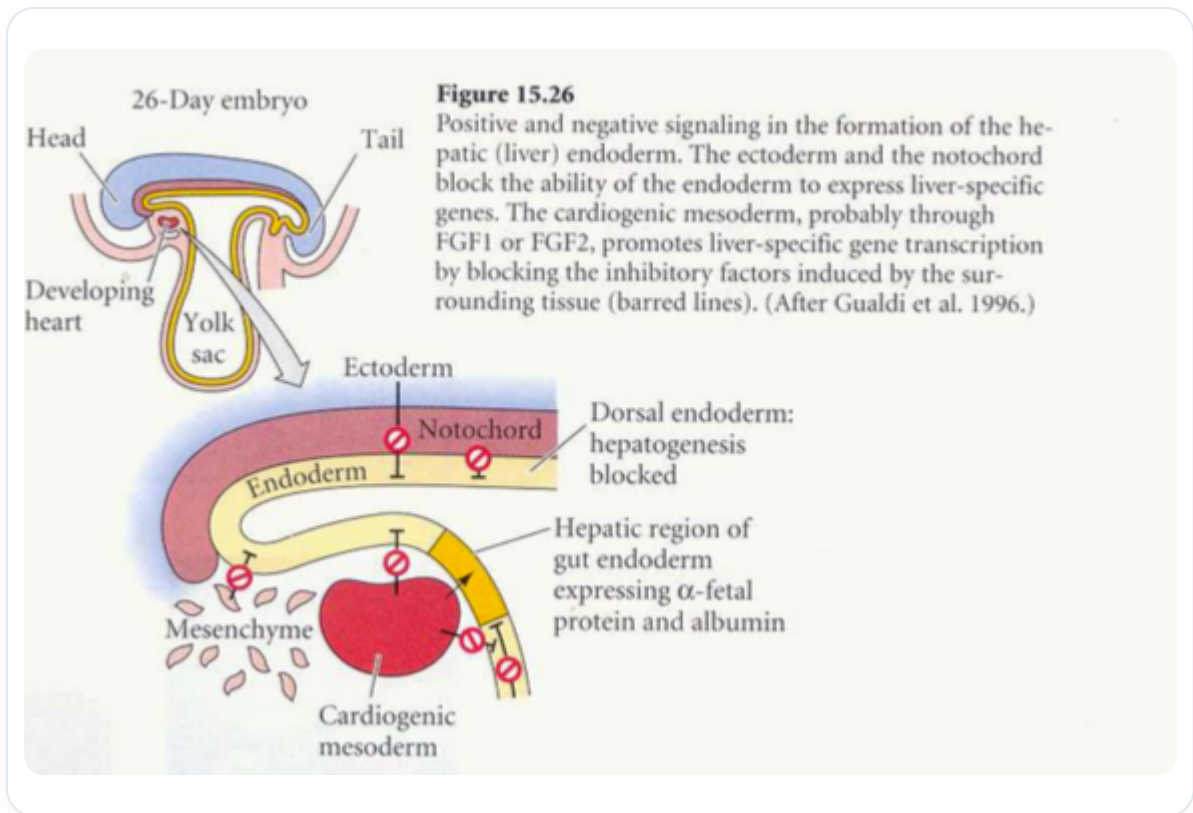


FIG. — Regulated embryonic hepatogenesis

Dynamically, the influence of **brain** development is decisive. It is through this influence that the cardiac structure develops, leading to a continuous flow of fluid towards the heart. This phenomenon creates a region of congestion in the primitive vascular system, which will form the basis of liver formation.

Brain development induces a forward flexion movement of the embryo, supported by a cardiac fulcrum. This movement causes congestion in a specific area, which will be the anlage of the liver. Thus, the development of the brain, heart, and diaphragm is interconnected and essential for liver formation.

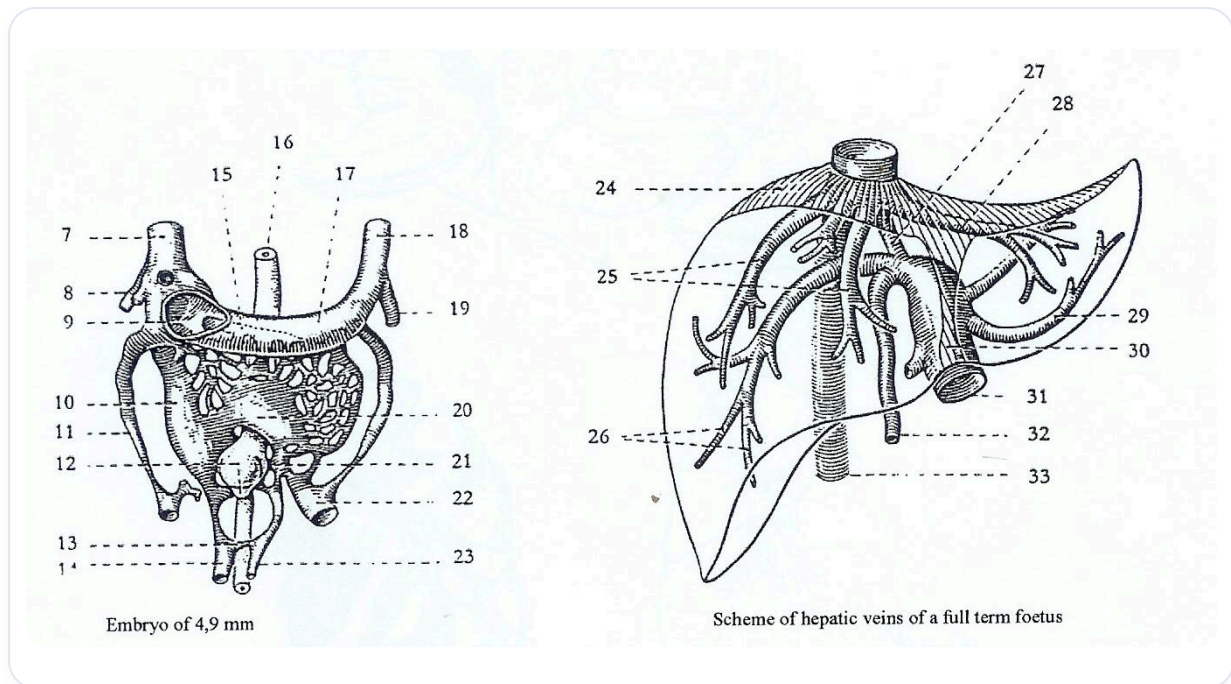


FIG. — *Embryo and fetal liver*

The liver emerges as an **expansion** of the fragmented digestive tube, consisting of a network of **bile ducts**. It represents an exocrine system that secretes substances into the intestinal lumen. The **mesodermal** structure of the liver, including its vascularisation, plays a fundamental role in its orientation and development.

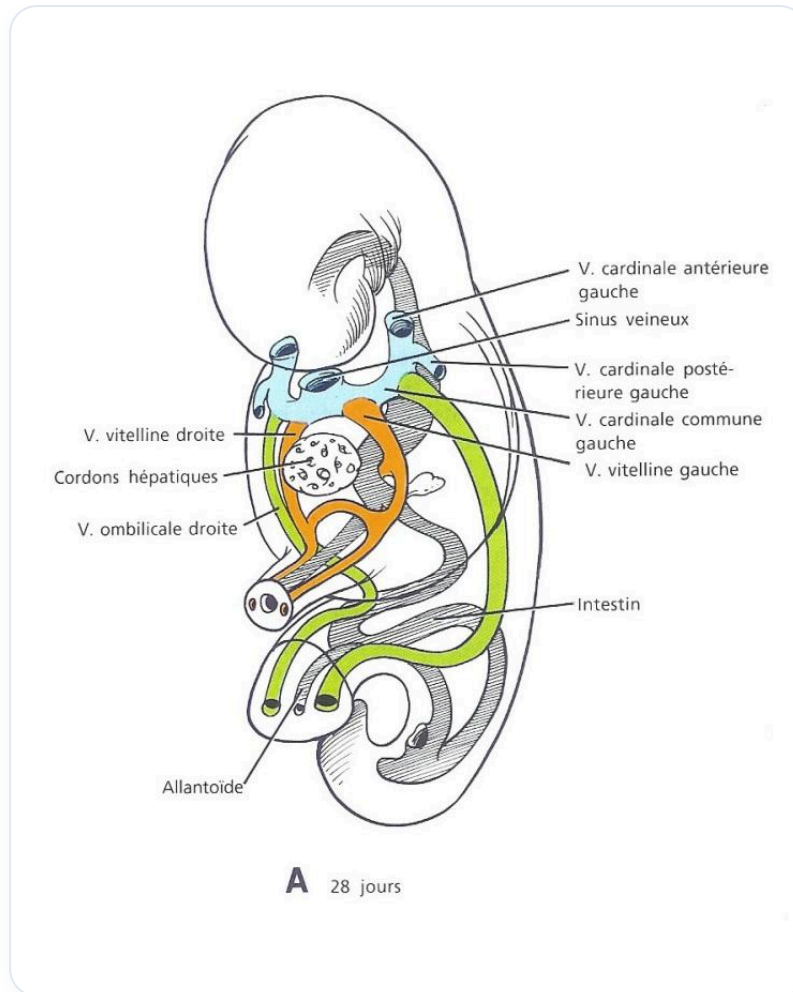


FIG. — 28-day old embryo venous system

It is important to remember that liver development is one of the main **fulcrums** for the organisation of the entire digestive tube. This process is orchestrated by the heart, itself influenced by the developing brain.

The liver develops from dissimilation products, which gives it its **cleansing** function. It is often described as the body's **purification plant**, working in synergy with the kidneys for waste elimination.

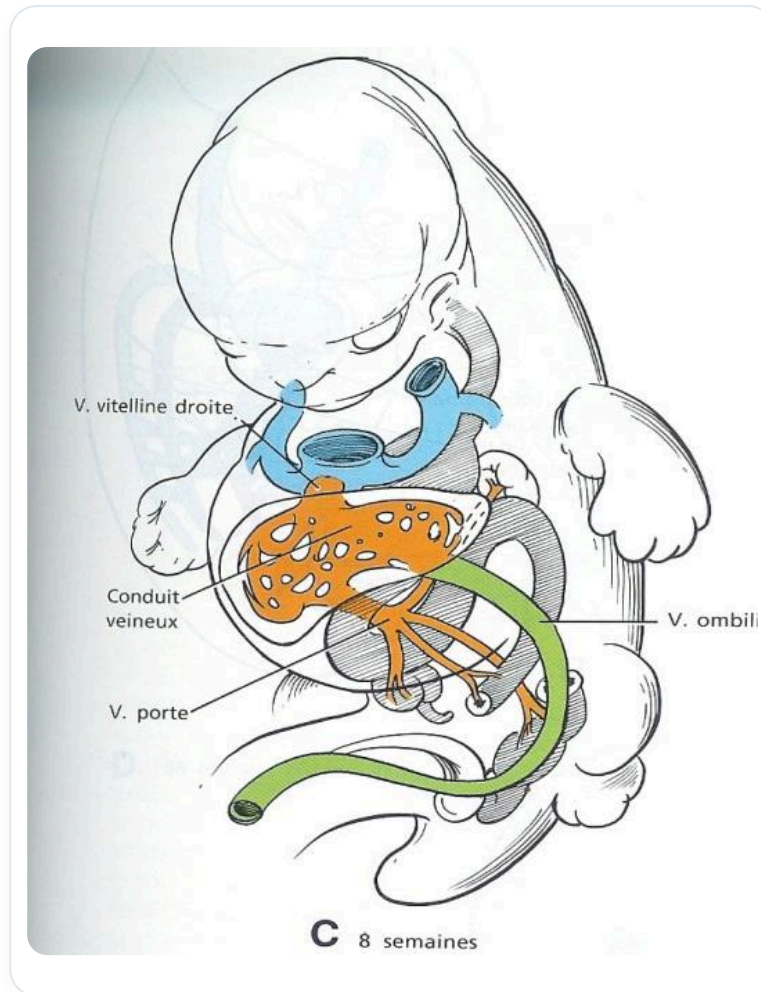


FIG. — Embryonic circulation 8 weeks

The vascular structure of the liver is complex, including the **vitelline veins** and the **umbilical veins**. These vessels form anastomoses that establish hepatic connections. The liver also contains the endothelial vitelline structure of the primitive vitelline vesicle.

The liver, as a central fulcrum, plays a crucial role in embryological balance. It is connected to the **pericardium**, **diaphragm**, and **falciform ligament**. These structures interact to create **synchronicity** in embryonic development.

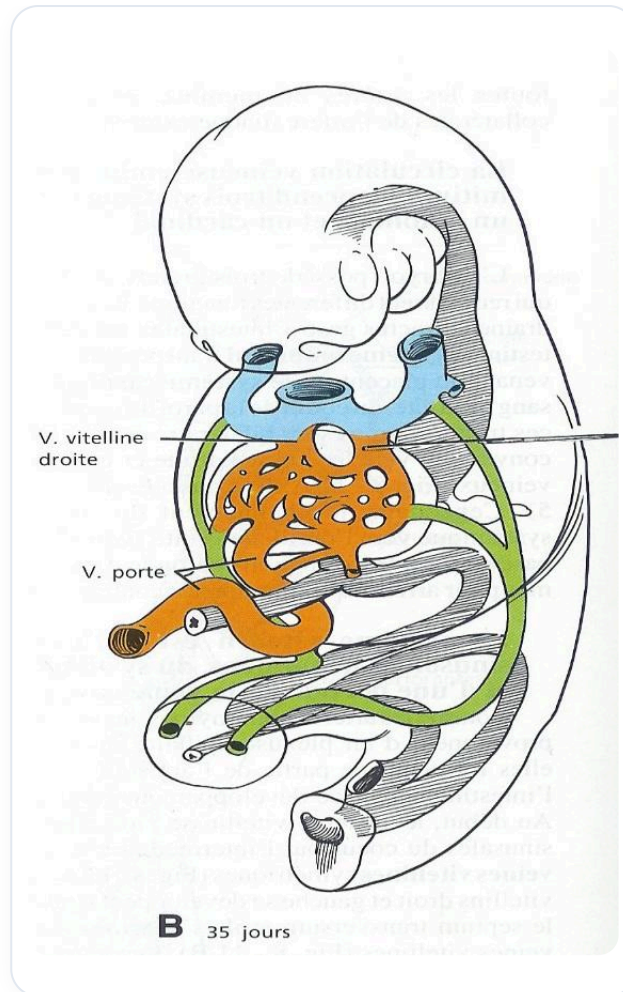


FIG. — *Embryo, venous circulation*

Hepatic development is synchronised with other structures, such as the **brain** and **cerebellum**. Tensions and movements in these common developmental spaces influence liver health and function.

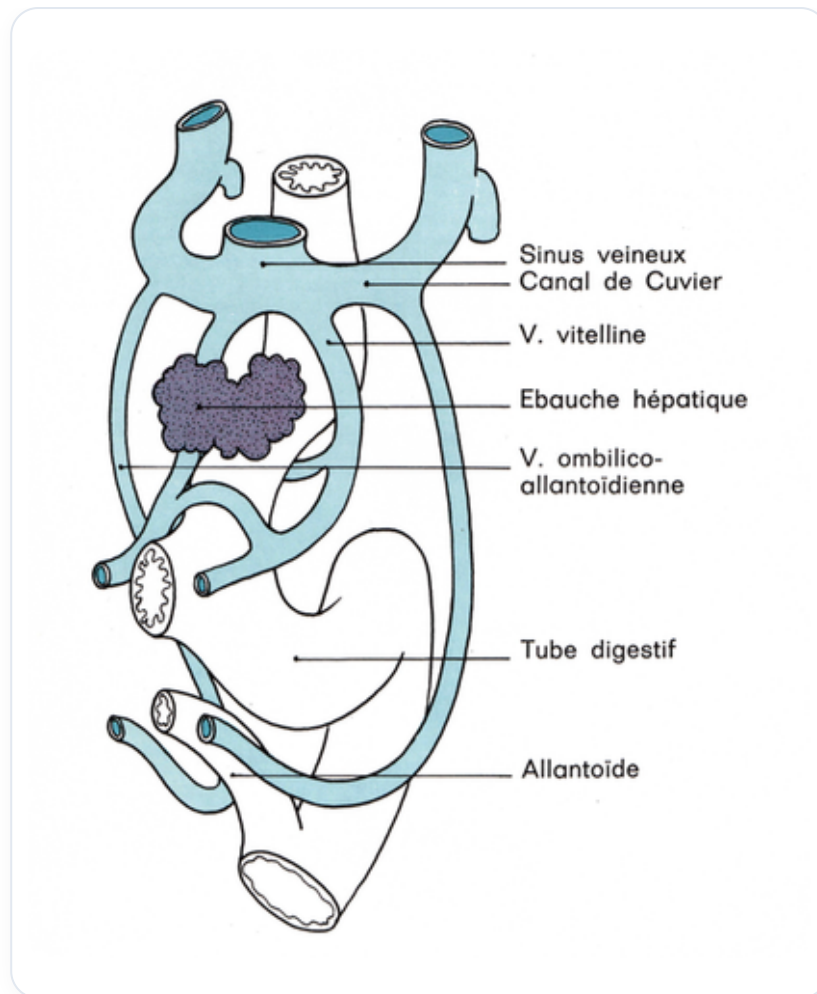


FIG. — Embryonic vascular development

Finally, it is essential to note that during the digestive phase, the liver receives a large amount of **venous blood** and little **arterial blood**. Conversely, during the night, circulation reverses, allowing the liver to regenerate at a cellular level.

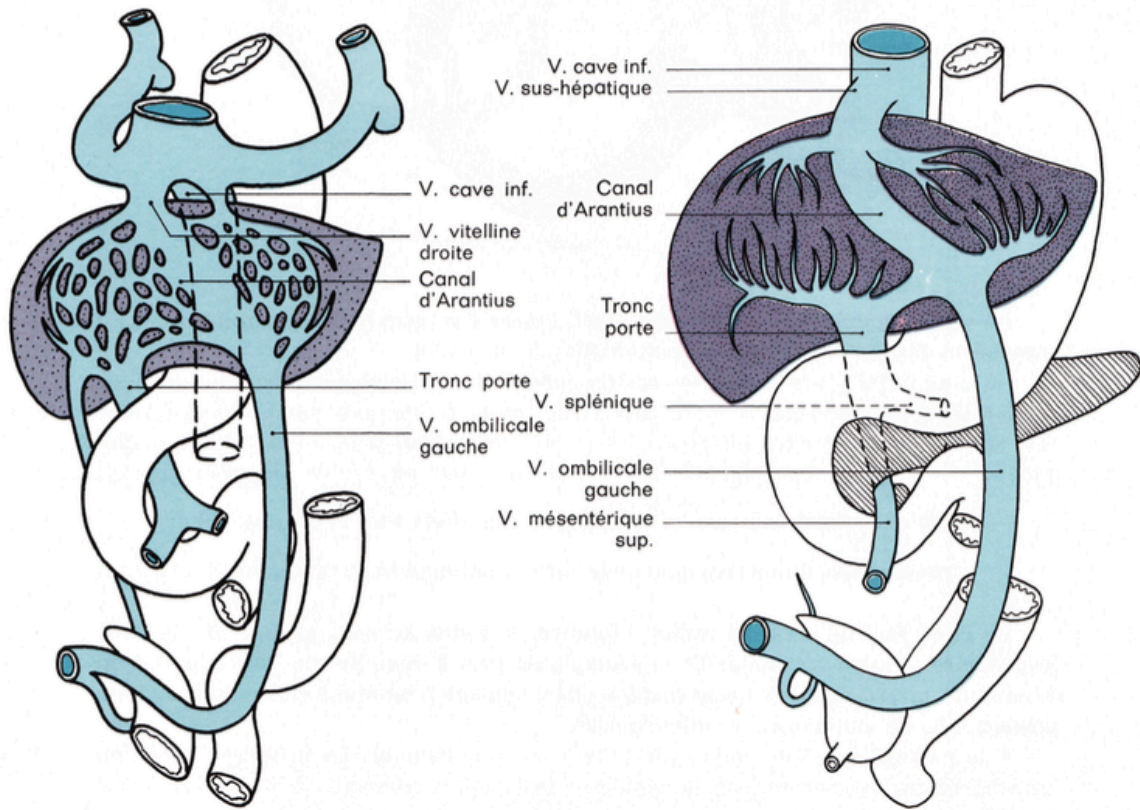


FIG. — Embryonic portal system

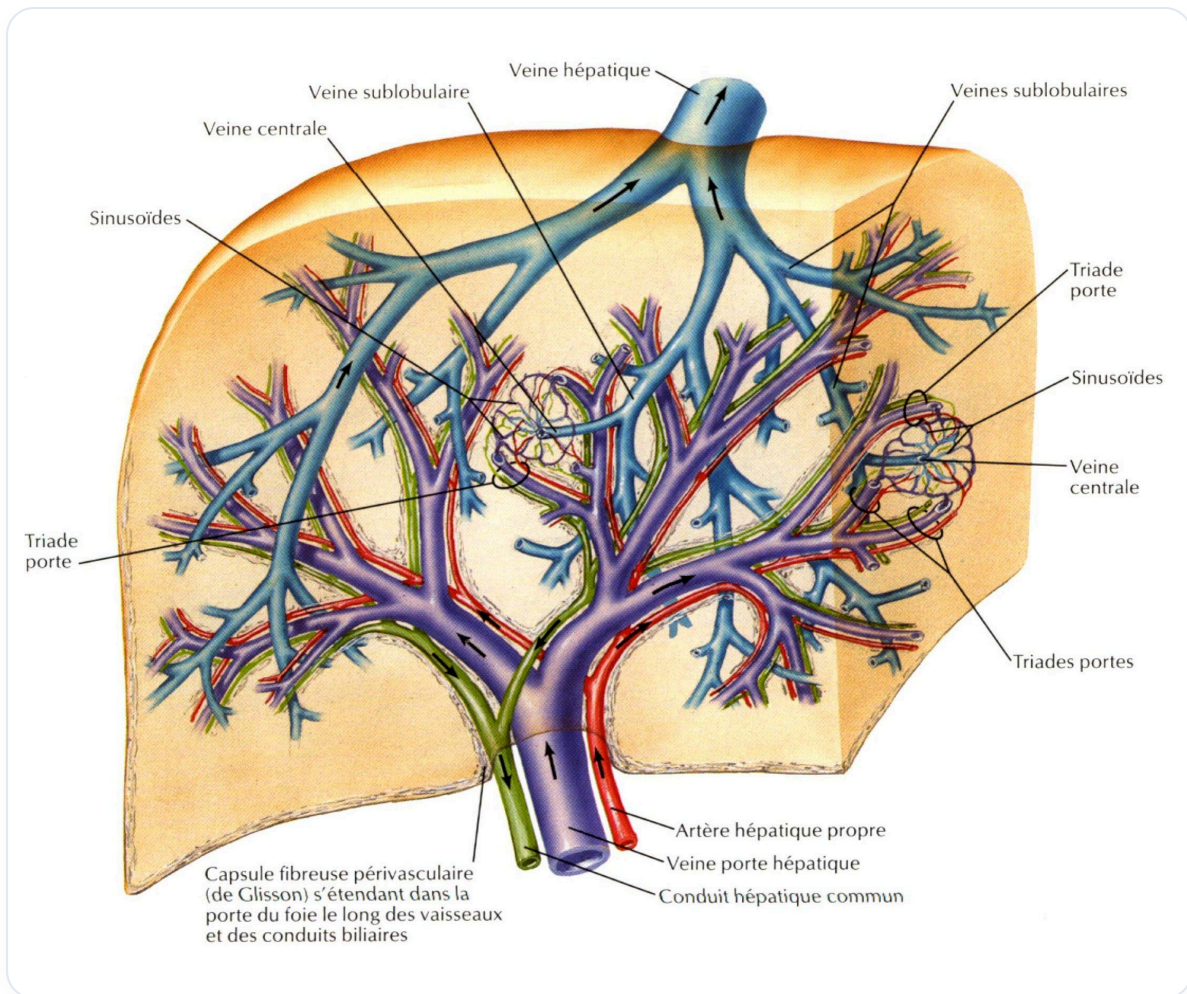
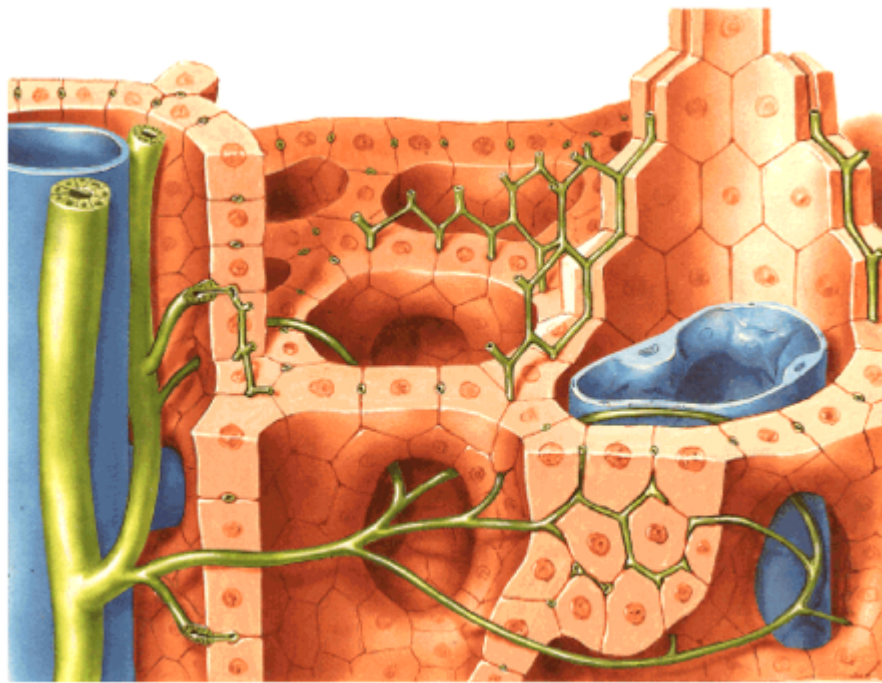


FIG. — Hepatic circulation



SISTEMA BILIAR INTRA-HEPÁTICO

FIG. — Intrahepatic biliary system